

NACA Airfoil Pressure Distribution

EGME 508: Advanced Inviscid Fluid Flow

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1 Abstract

This simulation sought to apply the method of source panels on an NACA 2412 airfoil to find the coefficient of pressure along the airfoil geometry. Matlab was used to generate the airfoil and to apply the method of panels onto said airfoil. The panel method is performed by approximating the flow as the sum of free stream and N panels (source or vortex panels). The same process was replicated on an NACA 0024 so the coefficient of pressure distributions could be compared.

It was determined that the coefficient of pressure along the lower profile of 2412 airfoil closely resembled that of the 0024 airfoil. But the upper profile produced a significantly different curve, and the difference in pressure between upper and lower curves is the reason that lift is generated on the airfoil.

2 Symbols and Units

Symbol	Measurement	Units
M	Max Camber	%
P	Max Camber Location	%
XX	Thickness	%
C	Chord	Units
α	Angle of Attack	° (degrees)
Y_U	Upper Coordinates	Units
Y_L	Lower Coordinates	Units
Y_{camber}	Camber Coordinates	Units
N_{ij}	Unit Normal Vector	Units
T_{ij}	Tangent Vector	Units
V_∞	Free Stream Velocity	m/s
V_i	Tangential Velocity	m/s
C_p	Coefficient of Pressure	Units

Note: Data is non-dimensionalized.

3 Methodology

The first step in using the panel method is to define an airfoil geometry using the NACA 4 digit airfoil and a computational method to develop a set of coordinates for a defined resolution based on a chosen NACA number. We will use these points to create our panels. After the geometry of the airfoil is defined, discretization of the airfoil surface is then defined. It's important to note that each panel has two endpoints. The goal of this discretization is to represent the airfoil geometry as a continuous series of straight line segments. Once the panel geometry is defined, control points (collocation points) are also defined. These reside at the center of each panel.

After general geometry is defined, the next step for the panel method is to calculate the unit normal N_{ij} and tangent T_{ij} vectors for each panel. These are used to apply boundary conditions and to calculate influence coefficients. Normal vectors point perpendicular to the panel surface and tangent vectors point along the panel surface. Boundary conditions are then defined with the free stream velocity being V_∞ . This can be imagined as the undisturbed flow velocity approaching the airfoil. Angle attack α is also defined, which is the angle between the free stream velocity and the chord line of the airfoil. Importantly, at each control point, enforce the boundary condition that the normal component of the velocity is zero. This condition ensures that there is no flow perpendicular to the surface of the airfoil.

The influence coefficients N_{ij} and T_{ij} represent the normal and tangential velocities induced at the control (collocation) points i by a unit source strength on panel j . A possible error could occur when $i = j$ as this represents the influence of a panel on itself treat these as a special case. Once accounted for, construct a system of linear equations based on the boundary conditions at each control point. The normal vector should be equal to zero as it represents zero flow perpendicular to the panel surface. Once made, solve the linear system for the unknown source strengths on each panel. These source strengths of course describe the flow distribution around the airfoil. Once the source strengths are known, calculate the tangent velocity at each control point by summing the contributions from each panel. This includes the free stream flow. This also known as the surface velocity.

Finally, the pressure distribution can be calculated. We apply Bernoulli's equation to find the pressure coefficients C_p which is given by $C_p = 1 - (\frac{V_i}{V_\infty})^2$. Where V_i and V_∞ represent the tangential and free stream velocities respectively. Use the tangential velocity at each control point to calculate the pressure coefficient at the given panel. Plot the given results along the surface of the airfoil to visualize how the pressure varies around the airfoil.

4 Results

The following figures were populated with data obtained from airfoiltools.com and the NACA Airfoil generation codes present in the appendix. The NACA 2412 and NACA 0024 airfoils were created and their pressure distributions were obtained.

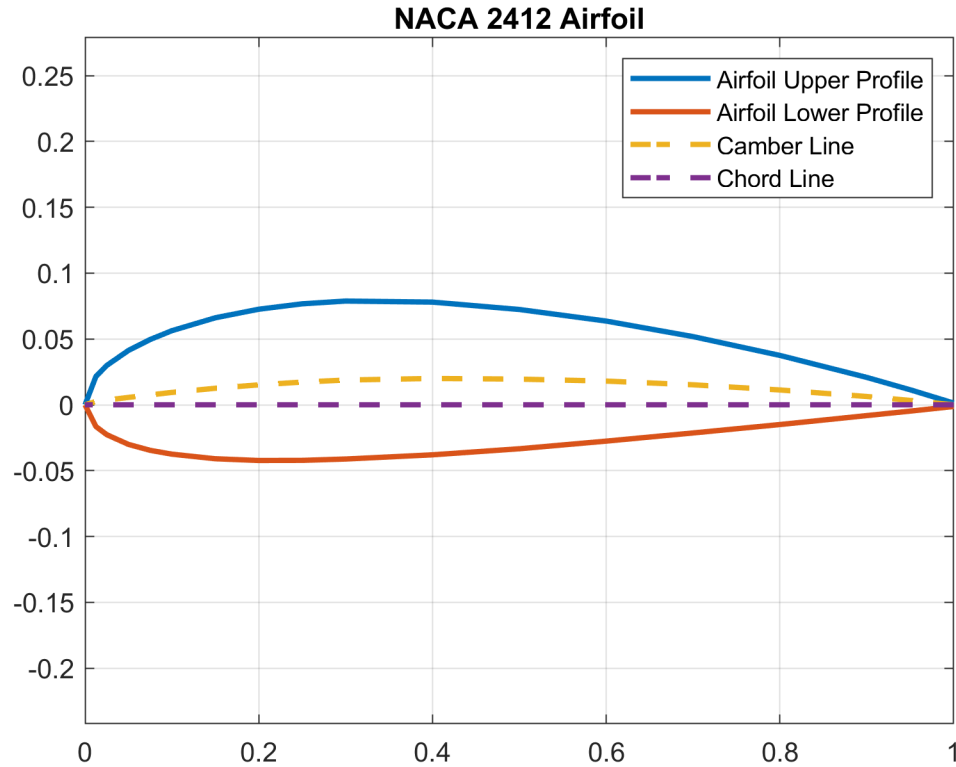


Figure 1. Tip to Tip Chord

Figure 1 represents the geometry of the NACA 2412 airfoil. One can see that the lower profile has a lesser magnitude compared to the upper profile due to the presence of the camber line. By having this asymmetric airfoil, lift can be generated when $\alpha = 0$. This means that there must be a difference in pressure distributions between the top and bottom profiles. This difference is illustrated further in Figure 3.

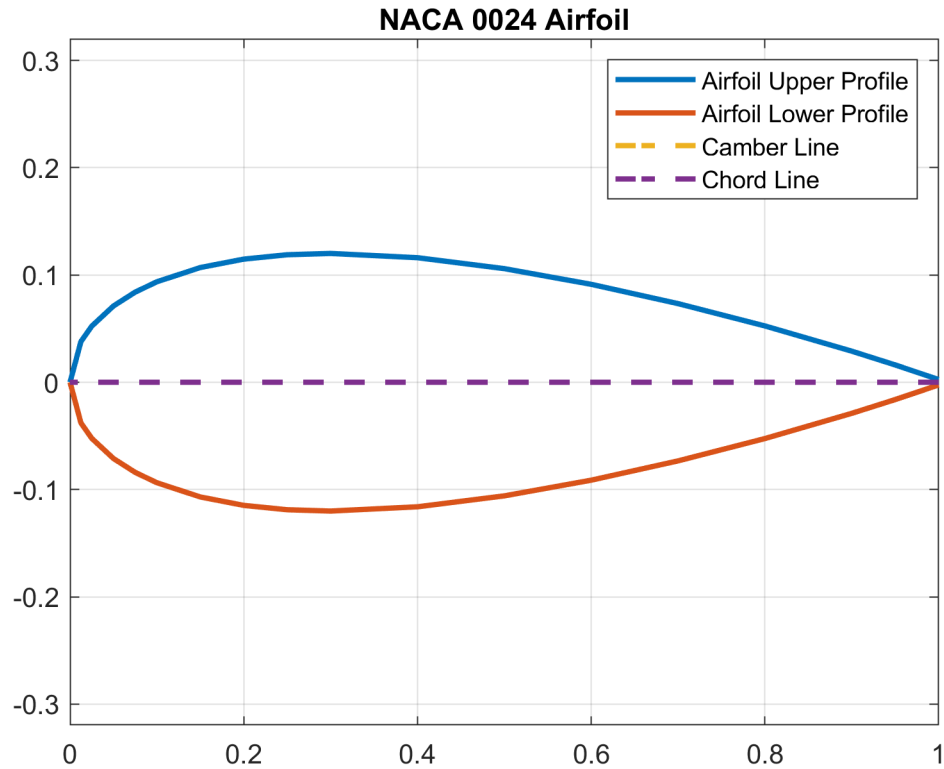


Figure 2. Tangent Line Chord

Figure 2 represents the geometry of the symmetric NACA 0024 airfoil. Unlike its asymmetric counterpart there is no camber line, which leads to the camber line and chord line being identical. The magnitude of both upper and lower profiles remain the same. Therefore, the measurements on either half of the geometry will be the same for pressure and velocity.

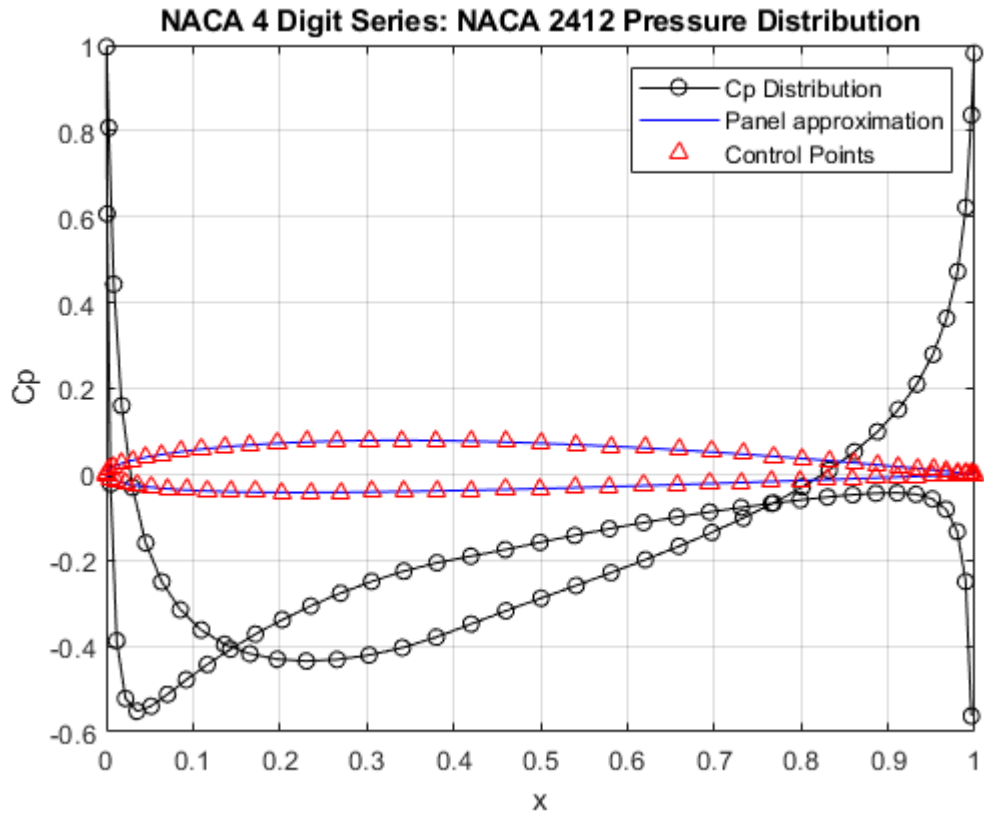


Figure 3. Chord Comparison

Figure 3 shows the pressure distribution of a NACA 2412 airfoil. Two pressure distribution lines are generated to account for the asymmetry. The more concave curve is the pressure distribution for the bottom half of the airfoil, whereas the more convex curve is for the top half of the airfoil. The pressure difference between the two profiles allows lift to be generated!

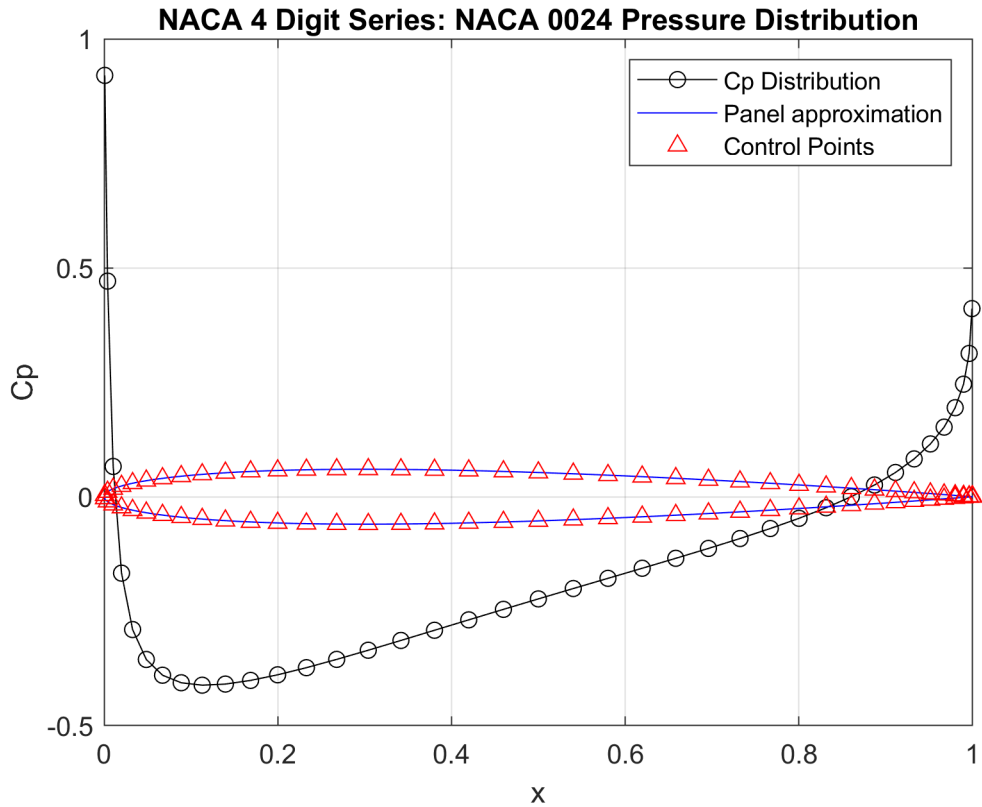


Figure 4. Chord Comparison

Figure 4 reveals the pressure distribution in the NACA 0024. Since this airfoil is symmetric, only one pressure distribution is revealed (pressure on the top and bottom are identical). Towards the leading edge, pressure is relatively low, while at the trailing edge it begins to rise quickly. Because the upper and lower profile pressure distributions are identical, this airfoil is unable to generate lift.

5 Discussion and Conclusion

The results from the Source Panel Method indicate the approximation of the airfoil geometry and control points were accurate in determining the coefficient of pressure for the NACA 2412 airfoil. The first step to achieve this was to generate the airfoil data using MATLAB. Data was generated for $N=40$, meaning there was going to be 40 points for each half of the airfoils; upper and lower. Data was then manipulated so that it could be passed on to the Source Panel function, which calculated the necessary components of velocity to calculate the source strengths and coefficient of pressure. The Source Panel method calculated the unit normal and tangential vectors on each of the panels and used the initial data to create the center points; X_C and Y_C . With these unit vectors, the perpendicular and tangential velocities, N_{ij} and T_{ij} , were calculated, thus leading to the pressure distribution. The coefficient of pressure curve is a consequence of the source panel method meaning that every defined panel along the airfoil influences every other panel of the airfoil. These panels were assumed to be a source at the center points influencing each other. The significance of the curves, as seen in Figure 3 and Figure 4, is that a pressure distribution can be approximated by knowledge of the free stream velocity and the airfoil geometry. The pressure variation on the 2412 airfoil varies depending on what side of the surface the calculation is being conducted, which is what is to be expected by the airfoil meant to be experiencing lift. The 0024 airfoil simply follows a singular path on the pressure distribution curve due to its symmetry. Its pressure also varies almost linearly for most of the body of the airfoil, which is also to be expected on a free stream. While the 2412 airfoil experiences less pressure on the upper surface between approximately 0.15 to 0.75, causing the pressure differential. A pressure differential also allows for lift to be generated! The limitations of this source panel method is going to be on the computation of the influence coefficients when N increases to a large number. For the purpose of approximating the pressure distribution on the airfoil, the source panel method shows convincing results.

This project shows the feasibility of using the method of source panels on various airfoils to compare their respective coefficients of pressure and therefore the lift they produce, thereby allowing us to evaluate the performance of different airfoils within the simulation. Doing so should also prove far cheaper and faster than running any physical experiments, while giving us meaningful information about various airfoils' ability to generate lift.

6 References

- [1] “NACA 2412.” NACA 2412 (NACA2412-IL), airfoiltools.com/airfoil/details?airfoil=naca2412-il. Accessed 3 Nov. 2024.
- [2] “NACA 0024.” NACA 0024 (NACA0024-IL), airfoiltools.com/airfoil/details?airfoil=naca0024-il. Accessed 3 Nov. 2024.
- [3] Chakraborty, Manash. (2015). A Computational Study on two horizontally close sequential airfoils to determine conjoined pressure distribution and aerodynamic influences on each other. 10.13140/RG.2.1.5041.4562.
- [4] Milne-Thomson, Louis Melville. “Theoretical Aerodynamics.Pdf.” Theoretical Aerodynamics, Dover Publications, Inc., 20 Jan. 2011, aerostarsolutions.wordpress.com/wp-content/uploads/2011/10/theoretical-aerodynamics.pdf.

7 Appendix A

7.1 MATLAB Code 1

```
% This script generates NACA 4 digits airfoil using the equations in
% Chapter 1 Section 5.6
%
clc; close all; clear parameters;
% Input Airfoil data ( remember NACA MPXX )
% M is the maximum camber divided by 100 (normalized by chord)
% P is the position of the maximum camber divide by 10 (normalized by chord)
% XX is the thickness divided by 100 (normalized by chord)
% Airfoil number
AirfoilAsk ='Enter The Airfoil Number in Raw Vector as [x x x x]: ';
Airfoil = input(AirfoilAsk);
NACA = Airfoil;

% setting up the increments on the x-axis
N=40; % number of increments
Beta_max=pi; Beta_min=0;
d_Beta=(Beta_max-Beta_min)/(N-1);
Beta=Beta_min:d_Beta:Beta_max;
% Equation for x
x=(1-cos(Beta))/2;

% Output what type of airfoil
if length(NACA) == 4
    disp(['NACA 4 Digit Series: NACA ',...
        num2str(NACA(1)) num2str(NACA(2)) num2str(NACA(3)) ...
        num2str(NACA(4))])
    if NACA(1) == 0 && NACA(2) == 0
        disp('Symmetric Airfoil')
    else
        Symm = 1;
        disp('Cambered Airfoil')
    end
end

% Airfoil equations
%
% Camber equations
% yc=M/P^2*(2Px-x^2) and dyc_by_dx=2M/P*(P-x) when 0<x<P
% yc=M/(1-P)^2*(1-2P+2Px-x^2) and dyc_by_dx=2M/(1-P)^2*(P-x) when P<x<1
M = NACA(1)/100;
P = NACA(2)/10;
T = str2double([num2str(NACA(3)),num2str(NACA(4))])/100;
yc=(x<P).*(M/P^2*(2*P*x-x.^2))+(x>=P).*(M/(1-P)^2*(1-2*P+2*P*x-x.^2));
dyc_by_dx=(x<P).*(2*M/P*(P-x))+(x>=P).*(2*M/(1-P)^2*(P-x));

% Thickness equations
% yt=T/0.2*(a_0*x.^0.5+a_1*x+a_2*x.^2+a_3*x.^3+a_4*x.^4)
% values of coefficients
% a_0=0.2969
% a_1=-0.126
% a_2=-0.3516
% a_3=0.2843
% a_4=-0.1015

a_0=0.2969;
a_1=-0.126;
a_2=-0.3516;
a_3=0.2843;
% a_4=-0.1015;
a_4=-0.1036; % closed trailing edge
yt=T/0.2*(a_0*x.^0.5+a_1*x+a_2*x.^2+a_3*x.^3+a_4*x.^4);
```

```

% Calculating the angle theta
theta=atan(dyc_by_dx);

% Calculating upper and lower surfaces
xu=x-yt.*sin(theta);
yu=yc+yt.*cos(theta);
xl=x+yt.*sin(theta);
yl=yc-yt.*cos(theta);

% Plots
figure(1)
hold on
grid on
% Plotting cambered airfoil
if Symm == 1
    plot(xu,yu,xl,yl,'LineWidth',1.5,'color','b')
    plot(x,yc,'--','LineWidth',1,'color','r')
    title(['NACA 4 Digit Series: NACA ', num2str(NACA(1))...
          num2str(NACA(2)) num2str(NACA(3)) num2str(NACA(4))])
    xlabel('x/c')
    ylabel('y/c')
    pbaspect([100 10 1])

% Plotting symmetric airfoil
else
    xu=x;
    xl=x;
    yu=yt/2;
    yl=-yt/2;
    plot(xu,yu,xl,yl,'LineWidth',1.5,'color','b')
    plot(x,yc,'--','LineWidth',1,'color','r')
    title(['NACA 4 Digit Series: NACA ', num2str(NACA(1))...
          num2str(NACA(2)) num2str(NACA(3)) num2str(NACA(4))])
    xlabel('x/c')
    ylabel('y/c')
    pbaspect([100 10 1])
end

%% This is the NACA 0024 offsets:
% Read the excel file where the panels have been identified.
% The Source_Panel function returns the N and T functions along with the
% n_hat and t_hat
%

% Reading airfoil data from an excel sheet
%data=readmatrix('NACA 0024 Airfoil.xlsx','Range','A7:B41');
%load('2412d.mat')
xu = transpose(xu);
xl = transpose(xl);
yu = transpose(yu);
yl = transpose(yl);
% Manipulation of the data
XP=[flip(xu);xl(2:end)]; YP=[flip(yu);yl(2:end)]; N = length(XP);
% Calling the panel source flow function
[AN,AT,XC,YC,NHAT,THAT] = Source_Panel(XP,YP,N);
% Setting the boundary conditions
uinfty = 1;
b=zeros(N,1);
for j=1:N
    b(j,1) = uinfty*NHAT(j,1);
end
% Solution of simultaneous algebraic equations
Sources = AN\b;
% Calculating tangential velocities
ut = AT*Sources - THAT(:,1);
% Coefficient of pressure
figure(2)

```

```

cp = 1 - ut.^2;
plot(XC(2:end-1),cp(2:end-1),'-ok','LineWidth',1)
hold on
plot(XP,YP,'k','LineWidth',1)
xlabel('X'),ylabel('Cp')
title(['NACA 4 Digit Series: NACA ', num2str(NACA(1))...
      num2str(NACA(2)) num2str(NACA(3)) num2str(NACA(4)), ' Pressure Distribution'])
grid on;

function [AN,AT,XC,YC,NHAT,THAT] = Source_Panel(XP,YP,N)
% Influence coefficients for source distribution over a symmetric body.
% AN is the Nij matrix and AT is the Tij matrix in Eqn (51).
for J = 1:N
    if J==1
        XPL = XP(N);
        YPL = YP(N);
    else
        XPL = XP(J-1);
        YPL = YP(J-1);
    end
    XC(J) = 0.5*(XP(J) + XPL); % Panel Center Coordinates
    YC(J) = 0.5*(YP(J) + YPL);
    S(J) = sqrt( (XP(J) - XPL)^2 + (YP(J) - YPL)^2 ); % Panel length
    THAT(J,1) = (XP(J) - XPL)/S(J);
    THAT(J,2) = (YP(J) - YPL)/S(J);
    NHAT(J,1) = - THAT(J,2);
    NHAT(J,2) = THAT(J,1);
end

%Calculation of the influence coefficients.
for I = 1:N
    for J = 1:N
        if I==J
            AN(I,J) = pi;
            AT(I,J) = 0;
        else
            DX = XC(I) - XC(J);
            DY = YC(I) - YC(J);
            XQ = DX*THAT(J,1) + DY*THAT(J,2);
            YQ = DX*NHAT(J,1) + DY*NHAT(J,2);
            VX = -0.5*( log( (XQ + S(J)/2 )^2 + YQ*YQ ) ...
                -log( (XQ - S(J)/2 )^2 + YQ*YQ ) );
            VY = -( atan2( (XQ + S(J)/2),YQ) - atan2((XQ - S(J)/2),YQ) );
            NTIJ = 0;
            NNIJ = 0;
            TTIJ = 0;
            TNIJ = 0;
            for K = 1:2
                NTIJ = NHAT(I,K)*THAT(J,K) + NTIJ;
                NNIJ = NHAT(I,K)*NHAT(J,K) + NNIJ;
                TTIJ = THAT(I,K)*THAT(J,K) + TTIJ;
                TNIJ = THAT(I,K)*NHAT(J,K) + TNIJ;
            end
            AN(I,J) = VX*NTIJ + VY*NNIJ;
            AT(I,J) = VX*TTIJ + VY*TNIJ;
        end
    end
end
end
end

```

7.2 MATLAB Code 2

```

clc, close all, clear, clear variables;

T1=readmatrix("NACA 2412.txt"); % For Project 2
T2=readmatrix("NACA 0024.txt"); % For Project 2

% Project 2 Data Fortmat
X1=T1(:,1); % X Data set Export
X2=T2(:,1); % X Data set Export
Yup1=T1(:,2); % Y Upper Data set Import
Yup2=T2(:,2); % Y Upper Data set Export
Ylow1=T1(:,3); % Y Lower Data set Export
Ylow2=T2(:,3); % Y Lower Data set Export

Ycamber1=(Yup1+Ylow1)/2; % Camber at each point in the data set
Ycamber2=(Yup2+Ylow2)/2; % Camber at each point in the data set
[Ycmax1,index_max_camb1]= max(Ycamber1); % Y Position of Max Camber and Position in data set
[Ycmax2,index_max_camb2]= max(Ycamber2); % Y Position of Max Camber and Position in data set
xmax1=X1(index_max_camb1); % X Position of Max Camber
xmax2=X2(index_max_camb2); % X Position of Max Camber
g1 = min(X1);
g2 = min(X2);

%% Plot 1
figure(1)
set(groot,'DefaultLineLineWidth',2)
plot(X1,Yup1) % Plotting Upper Half of Airfoil Profile
hold on
plot(X1,Ylow1) % Plotting Lower Half of Airfoil Profile
plot(X1,Ycamber1, '--') % Plotting Camber Line
plot(X1,g1+X1*0, '--') % Plotting Chord Line
hold off
legend('Airfoil Upper Profile', 'Airfoil Lower Profile', 'Camber Line', 'Chord Line')
xlim([0,1]) % Adjusting Aspect Ratio of plot
ylim([(min(Ylow1)-.2) (max(Yup1)+.2)])
title('NACA 2412 Airfoil')
grid on

%% Plot 2
figure(2)
set(groot,'DefaultLineLineWidth',2)
plot(X2,Yup2) % Plotting Upper Half of Airfoil Profile
hold on
plot(X2,Ylow2) % Plotting Lower Half of Airfoil Profile
plot(X2,Ycamber2, '--') % Plotting Camber Line
plot(X2,g2+X2*0, '--') % Plotting Chord Line
hold off
legend('Airfoil Upper Profile', 'Airfoil Lower Profile', 'Camber Line', 'Chord Line')
xlim([0,1]) % Adjusting Aspect Ratio of plot
ylim([(min(Ylow2)-.2) (max(Yup2)+.2)])
title('NACA 0024 Airfoil')
grid on

%% Results We Need
% Does ycamber max give the max camber as a percentage
Ycmaxper1=Ycmax1*100; % getting a % for max camber
Ycmaxper2=Ycmax2*100; % getting a % for max camber

% Max thickness
Thicc1 = Yup1 - Ylow1;
Thicc2 = Yup2 - Ylow2;
[Thicc_max1, Thicc_pos1] = max(Thicc1);
[Thicc_max2, Thicc_pos2] = max(Thicc2);
Thicc_max_per1 = Thicc_max1 * 100;
Thicc_max_per2 = Thicc_max2 * 100;
Thicc_max_loc1 = X1(Thicc_pos1);

```

```

Thicc_max_loc2 = X2(Thicc_pos2);

z1 = ['Max Camber as a Percentage of the Chord Length: ' num2str(Ycmaxper1) ' Percent'];
z2 = ['Location of Max Camber From Leading Edge: ' num2str(xmax1) ' Units'];
z3 = ['Max Thickness of Airfoil: ' num2str(Thicc_max_per1) ' Percent'];
z4 = ['Location of Max Thickness From Leading Edge: ' num2str(Thicc_max_loc1) ' Units'];
disp(z1)
disp(z2)
disp(z3)
disp(z4)

z5 = ['Max Camber as a Percentage of the Chord Length: ' num2str(Ycmaxper2) ' Percent'];
z6 = ['Location of Max Camber From Leading Edge: ' num2str(xmax2) ' Units'];
z7 = ['Max Thickness of Airfoil: ' num2str(Thicc_max_per2) ' Percent'];
z8 = ['Location of Max Thickness From Leading Edge: ' num2str(Thicc_max_loc2) ' Units'];
disp(z5)
disp(z6)
disp(z7)
disp(z8)

%% Panel Method Setup

% Manipulation of the data
XP=[X1;flipud(X1(1:end-1))];
YP=[Yup1;flipud(Ylow1(1:end-1))];
N = length(XP);
% Calling the panel source flow function
[AN,AT,XC,YC,NHAT,THAT] = Source_Panel(XP,YP,N);
b=zeros(N,1);
for j=1:N
    b(j,1) = NHAT(j,1);
end
Sources = AN\b;
ut = AT*Sources - THAT(:,1);

% Creating Panels
xtot = XP;
ytot = YP;
% Define panel resolution independently
panel_resolution = 11; % Adjust this to increase or decrease panel count
% Downsample xtot and ytot
XPan = xtot(1:round(length(xtot)/panel_resolution):end);
YPan = ytot(1:round(length(ytot)/panel_resolution):end);

% Plot 3: Coefficient of pressure
figure(3)
cp = 1 - ut.^2;
plot(XC(1:18),cp(1:18),'-ok')
hold on
plot(XC(18:end),cp(18:end),'-ob')
plot(X1,Yup1,'k')
plot(X1,Ylow1,'b')
plot(XPan,YPan,'r')
hold off
xlabel('X'),ylabel('Cp')
grid on;
title('NACA 2412 Pressure Distribution')

% Manipulation of the data
XP=[X2;flipud(X2(1:end-1))];
YP=[Yup2;flipud(Ylow2(1:end-1))];
N = length(XP);
% Calling the panel source flow function
[AN,AT,XC,YC,NHAT,THAT] = Source_Panel(XP,YP,N);
b=zeros(N,1);
for j=1:N
    b(j,1) = NHAT(j,1);
end

```



```

Sources = AN\b;
ut = AT*Sources - THAT(:,1);

% Creating Panels
xtot = [X2; flipud(X2)];
ytot = [Yup2; flipud(Ylow2)];
% Define panel resolution independently
panel_resolution = 11; % Adjust this to increase or decrease panel count
% Downsample xtot and ytot
XPan = xtot(1:round(length(xtot)/panel_resolution):end);
YPan = ytot(1:round(length(ytot)/panel_resolution):end);

% Plot 4: Coefficient of pressure
figure(4)
cp = 1 - ut.^2;
plot(XC,cp,'-ok ')
hold on
plot(X2,Yup2)
plot(X2,Ylow2)
plot(XPan,YPan,'r')
hold off
xlabel('X'),ylabel('Cp')
grid on;
title('NACA 0024 Pressure Distribution')

function [AN,AT,XC,YC,NHAT,THAT] = Source_Panel(XP,YP,N)
% Influence coefficients for source distribution over a symmetric body.
% AN is the Nij matrix and AT is the Tij matrix in Eqn (51).
for J = 1:N
    if J==1
        XPL = XP(N);
        YPL = YP(N);
    else
        XPL = XP(J-1);
        YPL = YP(J-1);
    end
    XC(J) = 0.5*(XP(J) + XPL); % Panel Center Coordinates
    YC(J) = 0.5*(YP(J) + YPL);
    S(J) = sqrt( (XP(J) - XPL)^2 + (YP(J) - YPL)^2 ); % Panel length
    THAT(J,1) = (XP(J) - XPL)/S(J);
    THAT(J,2) = (YP(J) - YPL)/S(J);
    NHAT(J,1) = - THAT(J,2);
    NHAT(J,2) = THAT(J,1);
end

%Calculation of the influence coefficients.
for I = 1:N
    for J = 1:N
        if I==J
            AN(I,J) = pi;
            AT(I,J) = 0;
        else
            DX = XC(I) - XC(J);
            DY = YC(I) - YC(J);
            XQ = DX*THAT(J,1) + DY*THAT(J,2);
            YQ = DX*NHAT(J,1) + DY*NHAT(J,2);
            VX = -0.5*( log( (XQ + S(J)/2 )^2 + YQ*YQ ) ...
                -log( (XQ - S(J)/2 )^2 + YQ*YQ ) );
            VY = -( atan2( (XQ + S(J)/2),YQ) - atan2((XQ - S(J)/2),YQ) );
            NTIJ = 0;
            NNIJ = 0;
            TTIJ = 0;
            TNIJ = 0;
            for K = 1:2
                NTIJ = NHAT(I,K)*THAT(J,K) + NTIJ;
                NNIJ = NHAT(I,K)*NHAT(J,K) + NNIJ;
                TTIJ = THAT(I,K)*THAT(J,K) + TTIJ;
                TNIJ = THAT(I,K)*NHAT(J,K) + TNIJ;
            end
        end
    end
end

```

```
        end
        AN(I,J) = VX*NTIJ + VY*NNIJ;
        AT(I,J) = VX*TTIJ + VY*TNIJ;
    end
end
end
end
```

7.3 MATLAB Code 3

```
% This script generates NACA 4 digits airfoil using the equations in
% Chapter 1 Section 5.6
%
clc; close all; clear parameters;clear all;clear;
% Input Airfoil data ( remember NACA MPXX )
% M is the maximum camber divided by 100 (normalized by chord)
% P is the position of the maximum camber divide by 10 (normalized by chord)
% XX is the thickness divided by 100 (normalized by chord)
% Airfoil number
AirfoilAsk ='Enter The Airfoil Number in Raw Vector as [x x x x]: ';
Airfoil = input(AirfoilAsk);
NACA = Airfoil;

% setting up the increments on the x-axis
%This will determine the approximation of the airfoil geometry
N=40; % number of increments
Beta_max=pi; Beta_min=0;
d_Beta=(Beta_max-Beta_min)/(N-1);
Beta=Beta_min:d_Beta:Beta_max;
% Equation for x
x=(1-cos(Beta))/2;
Symm = 0;
% Output what type of airfoil
if length(NACA) == 4
    disp(['NACA 4 Digit Series: NACA ',...
        num2str(NACA(1)) num2str(NACA(2)) num2str(NACA(3)) ...
        num2str(NACA(4))])
    if NACA(1) == 0 && NACA(2) == 0
        disp('Symmetric Airfoil')
    else
        Symm = 1;
        disp('Cambered Airfoil')
    end
end

% Airfoil equations
%
% Camber equations
%  $y_c = M/P^2 * (2Px - x^2)$  and  $dy_c/dx = 2M/P^2 * (P - x)$  when  $0 < x < P$ 
%  $y_c = M/(1-P)^2 * (1-2P+2Px - x^2)$  and  $dy_c/dx = 2M/(1-P)^2 * (P - x)$  when  $P < x < 1$ 
M = NACA(1)/100;
P = NACA(2)/10;
T = str2double([num2str(NACA(3)), num2str(NACA(4))])/100;
y_c = (x < P) .* (M/P^2 * (2*P*x - x.^2)) + (x >= P) .* (M/(1-P)^2 * (1-2*P+2*P*x - x.^2));
dy_c_by_dx = (x < P) .* (2*M/P^2 * (P - x)) + (x >= P) .* (2*M/(1-P)^2 * (P - x));

% Thickness equations
%  $y_t = T/0.2 * (a_0 * x.^{0.5} + a_1 * x + a_2 * x.^2 + a_3 * x.^3 + a_4 * x.^4)$ 
% values of coefficients
a_0=0.2969
a_1=-0.126
a_2=-0.3516
a_3=0.2843
a_4=-0.1015

a_0=0.2969;
a_1=-0.126;
a_2=-0.3516;
a_3=0.2843;
% a_4=-0.1015;
a_4=-0.1036; % closed trailing edge
y_t=T/0.2*(a_0*x.^0.5+a_1*x+a_2*x.^2+a_3*x.^3+a_4*x.^4);

% Calculating the angle theta
theta=atan(dy_c_by_dx);
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% Calculating upper and lower surfaces
xu=x-yt.*sin(theta);
yu=yc+yt.*cos(theta);
xl=x+yt.*sin(theta);
yl=yc-yt.*cos(theta);

% Plots
figure(1)
hold on
grid on
% Plotting cambered airfoil
if Symm == 1
    plot(xu,yu,xl,yl,'LineWidth',1.5,'color','b')
    plot(x,yc,'--','LineWidth',1,'color','r')
    title(['NACA 4 Digit Series: NACA ', num2str(NACA(1))...
          num2str(NACA(2)) num2str(NACA(3)) num2str(NACA(4))])
    xlabel('x/c')
    ylabel('y/c')

% Plotting symmetric airfoil
else
    xu=x;
    xl=x;
    yu=yt/2;
    yl=-yt/2;
    plot(xu,yu,xl,yl,'LineWidth',1.5,'color','b')
    plot(x,yc,'--','LineWidth',1,'color','r')
    title(['NACA 4 Digit Series: NACA ', num2str(NACA(1))...
          num2str(NACA(2)) num2str(NACA(3)) num2str(NACA(4))])
    xlabel('x/c')
    ylabel('y/c')
end

% Save the variables on current folder for use
% with the source panel method
save('2412d','xl','xu','yl','yu')

%
% This is the NACA 2412 offsets:
% Read the excel file where the panels have been identified.
% The Source_Panel function returns the N and T functions along with the
% n_hat and t_hat
%

% clc; clear parameters; close all;clear all;clear;

% Reading airfoil data from a .mat file generated by
% NACA Generation code with N number of panels

load('2412d.mat')
xu = transpose(xu);
xl = transpose(xl);
yu = transpose(yu);
yl = transpose(yl);

% Manipulation of the data
% Data to be made into a single column vector
XP=[flip(xu);xl(2:end)]; YP=[flip(yu);yl(2:end)]; N = length(XP);

% Calling the panel source flow function
[AN,AT,XC,YC,NHAT,THAT] = Source_Panel(XP,YP,N);

% Setting the boundary conditions
uinfty = 1;
b=zeros(N,1);
for j=1:N
    b(j,1) = uinfty*NHAT(j,1);

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end

% Solution of simultaneous algebraic equations
Sources = AN\b;
% Calculating tangential velocities
ut = AT*Sources - THAT(:,1);

% Coefficient of pressure
cp = 1 - ut.^2;

figure(2)
plot(XC(2:end-1),cp(2:end-1),'-ok ',XP,YP,'b-')
hold on
plot(XC,YP,'r^')
xlabel(' x '),ylabel(' Cp ')
grid on;
title('NACA 4 Digit Series: NACA 2412 Pressure Distribution')
legend('Cp Distribution','Panel approximation','Control Points')

function [AN,AT,XC,YP,NHAT,THAT] = Source_Panel(XP,YP,N)
% Influence coefficients for source distribution over a symmetric body.
% AN is the Nij matrix and AT is the Tij matrix in Eqn (51).
for J = 1:N
    if J==1
        XPL = XP(N);
        YPL = YP(N);
    else
        XPL = XP(J-1);
        YPL = YP(J-1);
    end
    XC(J) = 0.5*(XP(J) + XPL); % Panel Center Coordinates
    YC(J) = 0.5*(YP(J) + YPL);
    S(J) = sqrt( (XP(J) - XPL)^2 + (YP(J) - YPL)^2 ); % Panel length
    THAT(J,1) = (XP(J) - XPL)/S(J);
    THAT(J,2) = (YP(J) - YPL)/S(J);
    NHAT(J,1) = - THAT(J,2);
    NHAT(J,2) = THAT(J,1);
end

%Calculation of the influence coefficients.
for I = 1:N
    for J = 1:N
        if I==J
            AN(I,J) = pi;
            AT(I,J) = 0;
        else
            DX = XC(I) - XC(J);
            DY = YC(I) - YC(J);
            XQ = DX*THAT(J,1) + DY*THAT(J,2);
            YQ = DX*NHAT(J,1) + DY*NHAT(J,2);
            VX = -0.5*( log( (XQ + S(J)/2 )^2 + YQ*YQ ) ...
                -log( (XQ - S(J)/2 )^2 + YQ*YQ ) );
            VY = -( atan2( (XQ + S(J)/2),YQ) - atan2( (XQ - S(J)/2),YQ) );
            NTIJ = 0;
            NNIJ = 0;
            TTIJ = 0;
            TNIJ = 0;
            for K = 1:2
                NTIJ = NHAT(I,K)*THAT(J,K) + NTIJ;
                NNIJ = NHAT(I,K)*NHAT(J,K) + NNIJ;
                TTIJ = THAT(I,K)*THAT(J,K) + TTIJ;
                TNIJ = THAT(I,K)*NHAT(J,K) + TNIJ;
            end
            AN(I,J) = VX*NTIJ + VY*NNIJ;
            AT(I,J) = VX*TTIJ + VY*TNIJ;
        end
    end
end
end

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end